VQC ON SUPERCONDUCTING COMPUTERS

Presentation subtitle

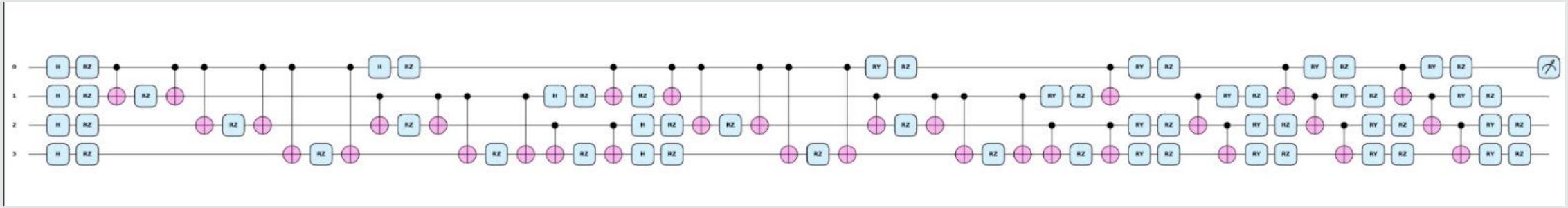
QUANTUM MACHINE LEARNING

Succesive Halving

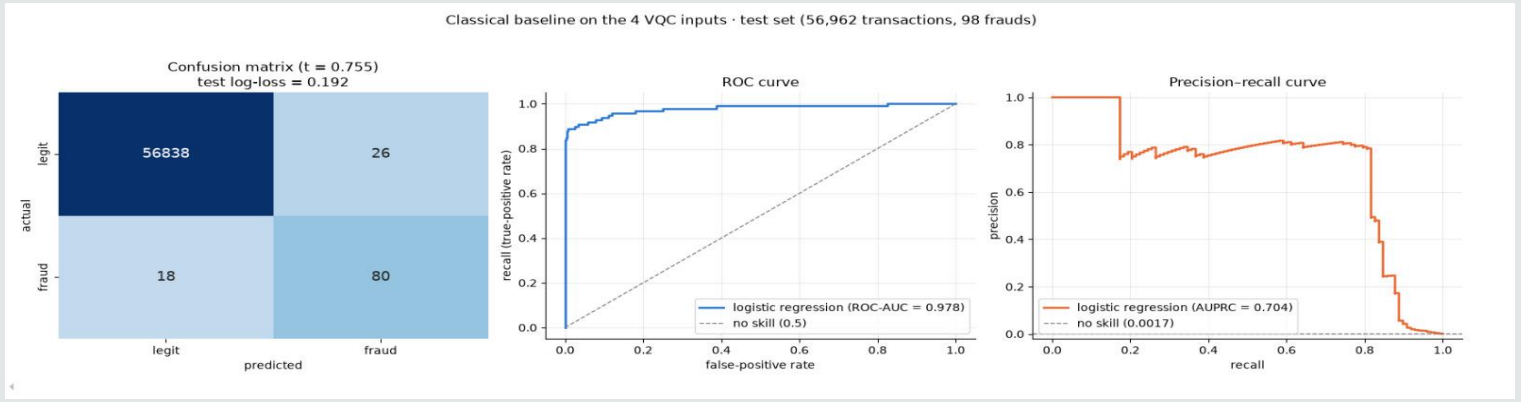
VQC

Dynamical Decoupling

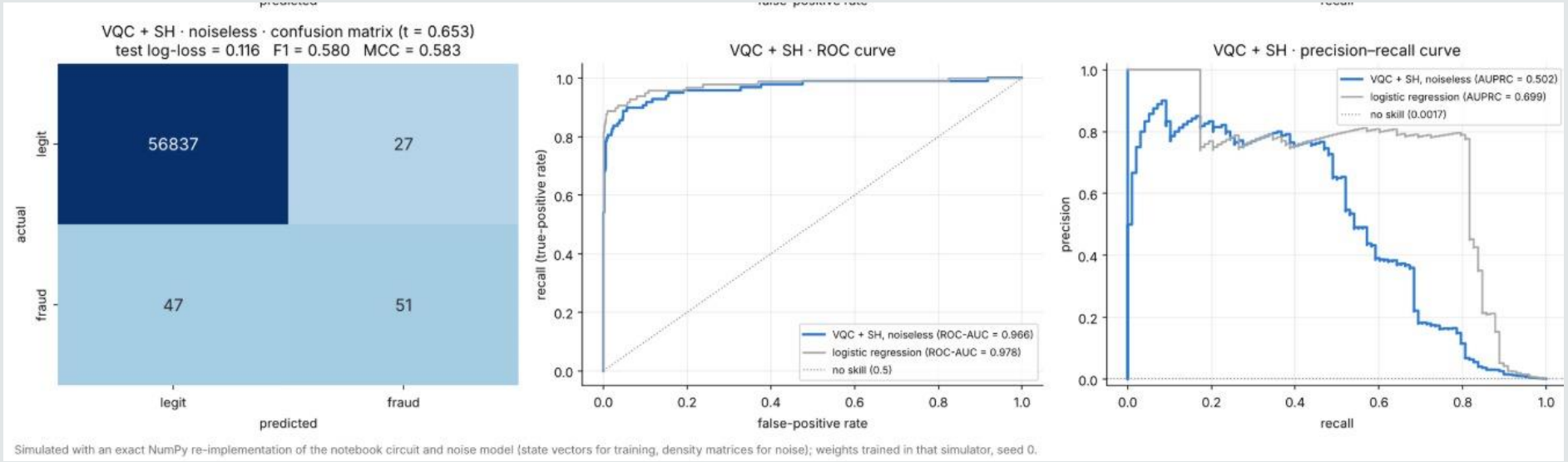
THE CIRCUIT



EFFICIENCY GRAPH

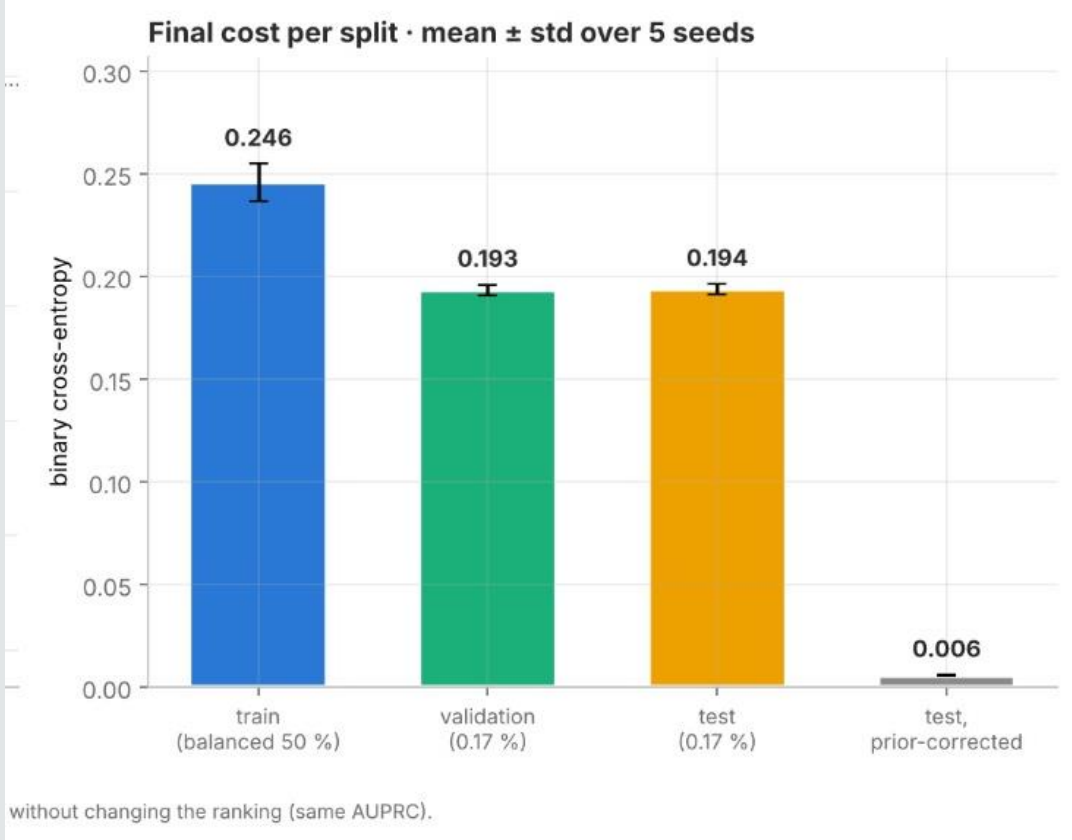


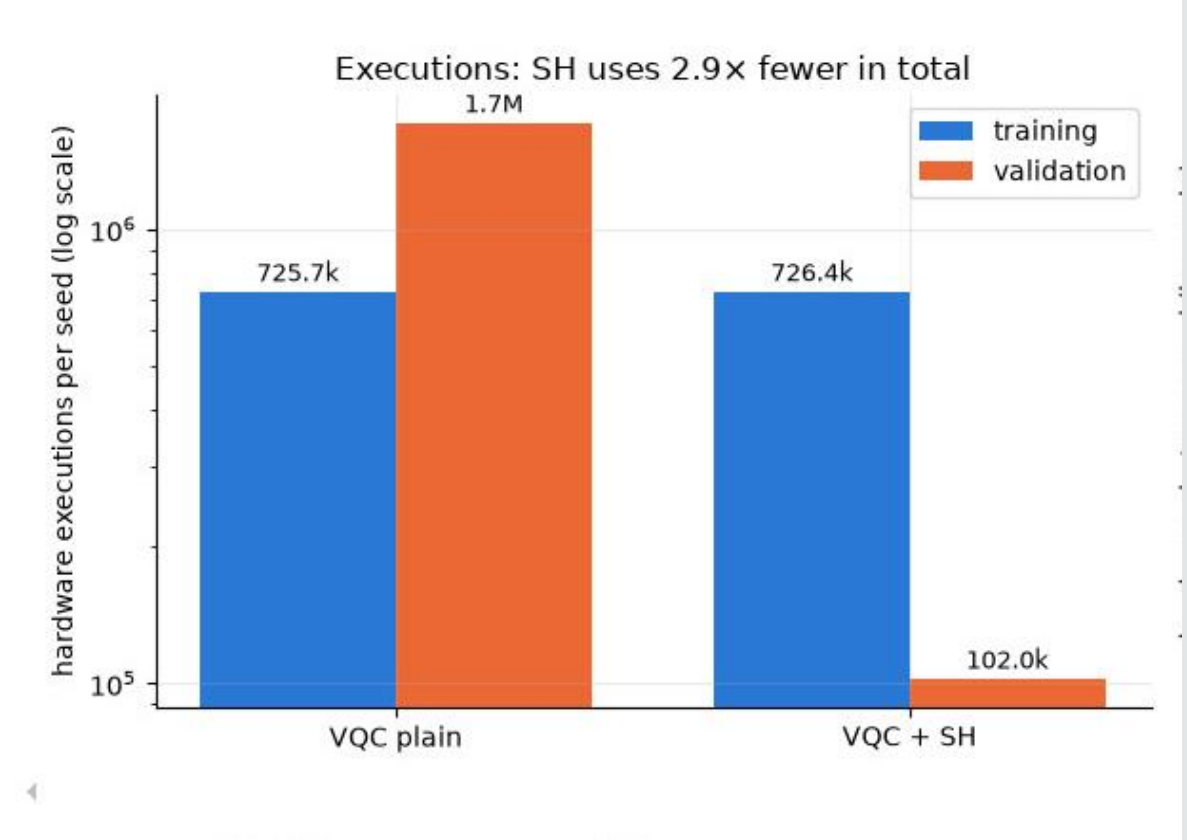
Classical ml



QML

COST GRAPH


Final cost per split · mean ± std over 5 seeds
binary cross-entropy
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.246
0.193
0.194
0.006
train
(balanced 50 %)
validation
(0.17 %)
test
(0.17 %)
test,
prior-corrected
without changing the ranking (same AUPRC).


Executions: SH uses 2.9× fewer in total
hardware executions per seed (log scale)
10^6
10^5
1.7M
725.7k
726.4k
102.0k
training
validation
VQC plain
VQC + SH

THANK YOU FOR YOUR ATTENTION!